

robvernonfirebaseedit

Travel Radio Australia, and we are here at the ANZ DMC, the Disaster Management Conference on the Gold Coast, and I have with me Rob Vernon from FireForesight. It's a tremendous company, it has amazing technology, which it predicts bushfires and actually helps in bushfire scenarios. Rob, welcome to the program.

Thank you, Geoff. Tell us about what you do, Rob. Rob, bushfires in Australia, as we know, are a big problem, and you're right in the forefront of everything.

Yeah, so basically what we're doing is using artificial intelligence for good, and we go about that in a number of different ways, combining a number of different technologies, all around the purpose of detecting bushfires as early as possible, and the idea there being the sooner we can detect them, the sooner we can pass that information on to emergency managers, and the sooner they can respond. Now, this all computerised, I suppose you've built all this technology over a number of years. How did it all start? We've come out of basically the forest industry in Tasmania, we've been developing a number of technologies that all around when is the right time to actually use fire, so beneficial fire, planned and prescribed burning.

So technologies like soil moisture monitoring, fuel moisture monitoring, weather stations, and combining all of that information in real time so that land managers have an understanding of when are the conditions going to be right, so that it's safe to conduct burns, it's not too wet, so there's not going to be too much smoke put into the atmosphere. Where we went to after that was basically then rolling out cameras to watch that, situational awareness, roll forward again, putting artificial intelligence into the mix to be at a landscape level, looking for early signs of fire that shouldn't be there. So I've got a lot of momentum behind what we do at the moment, we're doing that in every state around the country, we've had great success at home, but we're also building our network in Thailand, I was recently in the United States and we're looking at taking over networks there as well.

So global ambition in terms of what we're trying to do, but all developed here in Australia. Well it's fantastic, you know you've got a really snazzy machine here, it's got a giant, as I describe it, it's got a giant telescopic tower on it. Tell us about this machine, it's a big trailer.

Yeah, so this is what we call Firebase, so typically we go on the tops of tall places, you know Telco infrastructure or existing fire lookouts. Firebase gives us the opportunity to move around and be mobile, so based on what the weather's going to do this week, or if we can only be in a location temporarily, this we can drag into place, basically then erect the mast, get the camera up about 10 metres and then just be mobile in terms of being on the lookout for bushfires. So giving us a great opportunity to be in essence where we think fire is going to be and then provide that real-time guidance into emergency managers.

Of course we know that AI is taking over everything at the moment and your whole ethos is based around AI, how is it helping you? Yes, I mean fundamentally we're using AI to deal with scale, so Australia is a very big place, there are a lot of cameras that have great vantage points of the world, but we don't have enough people to be 24 by 7 monitoring a landscape the size of Australia, so the AI gives us that opportunity to be doing this every minute across a country as big as Australia or potentially even the world, so dealing with scale, continually improving that and then blending in other sources of information as well. We're combining data from sensors that are smelling for smoke, we're pulling in information from satellites that are there for fire detection, weaving all that together. Ultimately where we're going as a business is actually we want to play a role in terms of suppressing or mitigating early start fires as well, so I was recently in Alaska and we were testing AI autonomously swarming 15 drones to go and put fires out, so that's where we're headed as an organisation and we would not be able to do any of that without artificial intelligence.

Describe these drones for us because they're obviously not the little 250 milligram drones that you buy in the shop. Correct, correct. Think a drone perhaps the size of a ute and a big quadcopter, ultimately carrying as much payload as we can.

Ultimately to put out a fire you need a lot of suppressing, a lot of water as an example, so our approach to that is swarming, using multiple drones to basically provide an opportunity to get some payload early onto a fire. So you know it's a long journey on this working with emergency managers on where is the appropriate use of drones in an emergency situation. The last thing we want to be doing is getting in the way of existing aerial firefighting efforts.

It's a long burn on this, pardon the pun, but it's very much finding the right role for where drones sit in this and how we augment existing arrangements. And I suppose these drones could lay a line of fire in a retardant too, very much like the fire break but you don't have to burn anything down, right? We're speaking to Rob Vernon from FireForesight, it's a fascinating area. Unfortunately climate change is leading us to an environment where fire is even more prevalent.

Australia, the two big ones, fire and flood. Globally, just look at the news, what's going on at the moment. North America is on fire, a lot of Europe is on fire.

Decided a few years ago as a business we wanted to have impact in the world and fire and getting ahead of fire and playing an effective role in fire is absolutely something we want to do globally. Well it's a fascinating area, Rob, and thanks for being with us on the program. Can you give us your website where people can have a look and find out a bit more? Yeah, so it's just fireforesight.com. You'll be able to jump on there and find out all sorts of information about how we do what we do.

You'll be able to see some videos of Firebase in action and follow along with some of our journey around the world as well. Thanks for being part of the program, Rob. Thanks, Jeff.

